

Position Control of a Two-Propeller Aeropendulum: A Comparison Between Voltage and Current Actuation

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Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to check the author instructions for the journal they are submitting to for word limits and if structural elements like subheadings, citations, or equations are permitted.

Keywords: keyword1, Keyword2, Keyword3, Keyword4

1 Introduction

The Introduction section, of referenced text [?] expands on the background of the work (some overlap with the Abstract is acceptable). The introduction should not include subheadings.

Springer Nature does not impose a strict layout as standard however authors are advised to check the individual requirements for the journal they are planning to submit to as there may be journal-level preferences. When preparing your text please also be aware that some stylistic choices are not supported in full text XML (publication version), including coloured font. These will not be replicated in the typeset article if it is accepted.

2 Mathematical modeling

The physical structure of the twin-rotor aeropendulum can be conceptually mapped as a pendulum system operating in the vertical plane. The rod is constrained to a single rotational degree of freedom around a fixed low-friction pivot. The primary dynamics are governed by Newton's Second Law for rotational systems, which establishes that the net angular acceleration is directly proportional to the sum of all external torques acting on the apparatus:

$$\sum \tau_i(t) = I_{\text{total}}\alpha(t) = I_{\text{total}}\ddot{\theta}(t) \quad (1)$$

where I_{total} represents the total mass moment of inertia of the rigid body about its fixed axis of

rotation, $\theta(t)$ is the angular displacement relative to the vertical equilibrium position, and $\ddot{\theta}(t)$ denotes the mechanical angular acceleration.

To systematically derive the governing differential equations, the system is subjected to a comprehensive force and torque balance analysis. The schematic representation of the spatial distribution of the applied forces, coordinate systems, and geometric constraints is illustrated in Fig. 1.

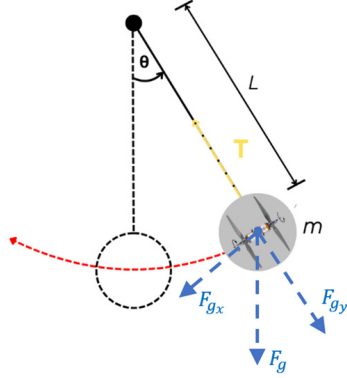


Fig. 1: Schematic diagram of the twin-rotor aeropendulum showing forces, dimensions, and angles.

As illustrated in Fig. 1, the structural setup is modeled as a mass m concentrated at the tip of a rigid rod of length L , deflected by an angle θ relative to the vertical equilibrium plane. The gravitational force $F_g = m_{\text{total}}g$ acting on the center of mass is decomposed into two orthogonal components: the radial force $F_{gy} = m_{\text{total}}g \cos \theta$, which acts along the longitudinal axis of the rod and is counterbalanced by the internal structural tension force T , and the tangential force $F_{gx} = m_{\text{total}}g \sin \theta$. This tangential component acts perpendicularly to the rod, consistently pulling the system back toward the stable vertical rest position ($\theta = 0^\circ$).

By taking the pivot as the rotational origin and applying the principle of superposition to all coincident perpendicular effects, the dynamic torque balance equation is expressed as:

$$\tau_m(t) + \tau_r(t) + \tau_a(t) = I_{\text{total}}\ddot{\theta}(t) \quad (2)$$

where $\tau_m(t)$ represents the active net propulsive torque generated by the twin-propeller assembly at the tip, defined by the operational difference between the front and rear actuators ($\tau_m(t) = \tau_{m1}(t) - \tau_{m2}(t)$). The term $\tau_r(t)$ is the gravitational restoring torque directly derived from the tangential force component shown in the schematic diagram, defined as $\tau_r(t) = -LF_{gx} = -Lm_{\text{total}}g \sin \theta(t)$, where the negative sign denotes its restorative nature opposing the angular displacement. Finally, $\tau_a(t)$ is the non-conservative aerodynamic drag torque that models the atmospheric fluid resistance opposing the motion of the rod through the air, given by:

$$\tau_a(t) = -b\dot{\theta}(t) \quad (3)$$

where b denotes the rotational viscous damping coefficient and $\dot{\theta}(t)$ represents the instantaneous angular velocity of the plant.

The structural resistance to change in its rotational state, quantified by the total moment of inertia (I_{total}) about the rotation axis, must also be parameterized. This value is determined by combining the continuous mass distribution of the homogeneous rigid rod and the discrete masses of the two actuators located at its tip. Assuming a uniform mass distribution along the rod of length L and mass m_{haste} , and modeling the two BLDC motors as point masses of value m_{motor} located at a distance L from the pivot, the total equivalent moment of inertia is established via superposition as:

$$I_{\text{total}} = L^2 \left(\frac{1}{3}m_{\text{haste}} + 2m_{\text{motor}} \right) \quad (4)$$

Once all individual torque components acting on the mechanical structure and the geometric parameters are explicitly defined, they can be directly substituted into the general mechanical balance represented in (2). Consequently, the full non-linear second-order ordinary differential equation that governs the twin-rotor aeropendulum dynamic behavior is obtained:

$$L^2 \left(\frac{1}{3}m_{\text{haste}} + 2m_{\text{motor}} \right) \ddot{\theta}(t) + b\dot{\theta}(t) + Lm_{\text{total}}g \sin \theta(t) = \tau_{m1}(t) - \tau_{m2}(t) \quad (5)$$

For the development of classical linear control architectures, the non-linear operating region can

be linearized around the stable vertical equilibrium position ($\theta = 0^\circ$). Utilizing the small-angle Taylor approximation ($\sin \theta \approx \theta$), the fundamental expression reduces to a linear ordinary differential equation. By assuming that the structural lumped parameters can be aggregated as:

$$J = L^2 \left(\frac{1}{3} m_{\text{haste}} + 2m_{\text{motor}} \right) \quad (6)$$

$$K_g = Lm_{\text{total}}g \quad (7)$$

where J denotes the total equivalent moment of inertia and K_g represents the gravitational stiffness coefficient. Substituting (6) and (7) into the linearized formulation simplifies the equation of motion to:

$$J\ddot{\theta}(t) + b\dot{\theta}(t) + K_g\theta(t) = \tau_{m1}(t) - \tau_{m2}(t) \quad (8)$$

Assuming zero initial mechanical conditions ($\theta(0) = 0$ and $\dot{\theta}(0) = 0$), the application of the Laplace transform to this model maps the dynamics directly into the complex frequency s -domain. Ultimately, the open-loop transfer function $G(s)$ that establishes the dynamic relationship between the net propulsive torque input and the resulting angular position output $\Theta(s)$ fits seamlessly within a single column line, being rigorously derived as:

$$G(s) = \frac{\Theta(s)}{T_{m1}(s) - T_{m2}(s)} = \frac{1}{Js^2 + bs + K_g} \quad (9)$$

3 Experimental platform and parameter identification

To evaluate the mathematical formulations and validate the proposed control strategies, a dedicated experimental testbed was developed and utilized in this work. The physical apparatus consists of a twin-rotor aeropendulum framework, illustrated in Fig. 2, designed for the study and benchmark of non-linear control techniques.

The mechanical assembly isolates the plant's movement to a single rotational degree of freedom within the vertical plane. The main structural components, geometric constraints, and the instrumentation infrastructure are detailed below.

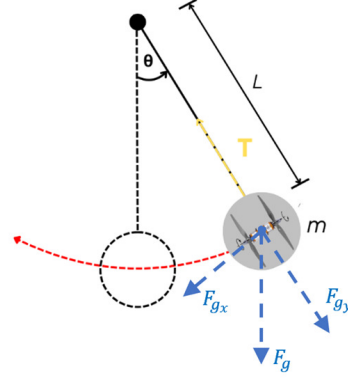


Fig. 2: General physical structure of the twin-rotor aeropendulum experimental platform.

3.1 Mechanical and Hardware Architecture

The structural support consists of a rigid frame that holds the primary rotational shaft. To minimize frictional losses and guarantee that the plant dynamics are predominantly driven by the active aerodynamic inputs, the mechanical assembly is coupled to the fixed frame through low-friction bearings. The dimensional design of the prototype is depicted in Fig. ??.

The primary physical and operational characteristics of the embedded subsystems are defined as follows:

- **Propulsion Subsystem:** Actuation is provided by two A2212/13T Brushless DC (BLDC) motors mounted back-to-back at the lower tip of a rigid aluminum rod ($L = 0.515$ m). Each motor is coupled to a $7 \times 4 \times 3$ polycarbonate tri-blade propeller. This assembly converts the digital control commands into active differential thrust forces perpendicular to the longitudinal axis of the rod.
- **Processing and Control Unit:** The computational core is driven by an ATmega2560 8-bit microcontroller (Arduino Mega platform). This unit is responsible for executing the real-time digital filtering routines, processing the control laws, and generating the independent high-resolution Pulse Width Modulation (PWM) signals dispatched to the Electronic Speed Controllers (ESCs). The control loop runs at a

fixed sampling period, and data communication with the host computer is established via a bidirectional USB/Serial interface.

- **Sensors and Interface Subsystems:** The angular position $\theta(t)$ is acquired through a high-precision rotary potentiometer attached directly to the main pivot axis, selected due to its high linearity and noise immunity. For manual operational testing and setpoint tuning without an ungrounded graphical user interface, an analog joystick module was integrated into the electronic bus.
- **Power Supply Unit:** To ensure voltage stability and satisfy the strict current demands during high-acceleration transients, the entire platform is powered by a regulated 10 V, 30 A switched-mode power supply (SMPS).

3.2 Lumped Parameter Verification

By crossing the physical dimensions from the technical specification with the nominal mass data ($m_{\text{haste}} = 0.210$ kg and $m_{\text{motor}} = 0.053$ kg), the total mass concentrated at the application radius is established as $m_{\text{total}} = 0.316$ kg. Under standard gravity ($g = 9.81$ m/s²), the lumped equivalent moment of inertia (J) and the gravitational stiffness gain (K_g), previously defined in Section 2, are analytically computed as:

$$J = L^2 \left(\frac{1}{3} m_{\text{haste}} + 2m_{\text{motor}} \right) \approx 0.0234 \text{ kg} \cdot \text{m}^2 \quad (10)$$

$$K_g = Lm_{\text{total}}g \approx 1.596 \text{ N} \cdot \text{m/rad} \quad (11)$$

To fully parameterize the dynamic model, the experimental calibration curves regarding the voltage-to-torque and current-to-PWM conversions—along with their respective linear regression fits—are addressed in the following subsections.

4 Input signal interpretation and actuation dynamics

In phenomenological modeling, the input is conventionally defined by the pure net aerodynamic

torque developed by the propulsion units. However, from a practical control systems perspective, the actuators cannot be directly driven by a torque command. Instead, they must be excited via voltage or current variables. This section establishes the mathematical and experimental formulations for two distinct actuation paradigms: voltage-based equivalent actuation and estimated current-based actuation. These formulations redefine the open-loop plant inputs, establishing the basis for the comparative analysis.

4.1 Voltage-Based Actuation Dynamics

The operation of Brushless DC (BLDC) motors relies on the electromagnetic interaction between the stator windings and the permanent magnet rotor. The fundamental torque generation is structurally governed by the motor torque constant (K_t), which linearly relates the mechanical torque to the armature current (i_a):

$$\tau(t) = K_t i_a(t) \quad (12)$$

noindent where K_t is the motor torque constant, which is inversely proportional to the motor velocity constant (K_v , specified by the manufacturer as 1000 RPM/V for the actuators utilized in this platform) according to the structural relation:

$$K_t = \frac{60}{2\pi K_v} \approx 0.00955 \text{ N} \cdot \text{m/A} \quad (13)$$

To establish a direct practical relationship between the applied terminal voltage (V) and the resulting torque (τ), this work proposes an experimental characterization based on steady-state equilibrium conditions. Under this framework, we propose that the armature current $i_a(t)$ can be correlated to the structural angular displacement $\theta(t)$ attained by the aeropendulum through an empirical proportionality coefficient K_{fi} :

$$i_a(t) = K_{fi} \theta(t) \quad (14)$$

Concurrently, we propose a secondary linear relationship to link the steady-state angular displacement to the equivalent average terminal voltage ($V(t)$) applied to the Electronic Speed Controllers (ESCs). This correlation is defined by the coefficient K_{fv} as follows:

$$\theta(t) = K_{fv} V(t) \quad (15)$$

By combining these proposed steady-state relations, the active torque produced by an isolated motor can be coupled to its terminal voltage input through a lumped equivalent voltage-to-torque gain ($K_{\text{equivalent}}$):

By substituting (15) into (14), and the result into the fundamental torque relation (12), the active torque produced by an isolated motor can be directly coupled to its terminal voltage input through a lumped equivalent voltage-to-torque gain ($K_{\text{equivalent}}$):

$$\tau(t) = K_t K_{fi} K_{fv} V(t) = K_{\text{equivalent}} V(t) \quad (16)$$

To parameterize this model for the bidirectional setup, systematic experimental calibration procedures were conducted. Specific Pulse Width Modulation (PWM) duty cycles were applied stepwise to the front and rear actuators. The corresponding armature current, equivalent terminal voltage, and steady-state angular deflections were recorded.

Linear regression algorithms were applied to the experimental curves, mapping the correlation parameters. The calculated gains K_{fi} , K_{fv} , and the resulting lumped $K_{\text{equivalent}}$ parameters for both propulsion units are consolidated in Table 1.

Table 1: Static calibration parameters for voltage actuation

Actuator	K_{fi} (A/°)	K_{fv} (°/V)	$K_{\text{equivalent}}$ (N · m/V)
Front	0,053252	8,26146	0,00420
Rear	-0,121940	-5,11253	0,00595

Thus, by applying the algebraic difference between the independent actuators, the net voltage-driven propulsive torque command is defined as $\tau_{m1}(t) - \tau_{m2}(t) = K_{\text{equivalent},1} V_1(t) - K_{\text{equivalent},2} V_2(t)$.

4.2 Current-Based Actuation Dynamics

Unlike the voltage paradigm where torque coupling is indirect and relies on multi-stage linear approximations—current-based actuation leverages the direct, uncoupled physical relationship

between armature current and torque expressed in (12). Actuating the plant via current provides a more linear control loop, as it inherently diminishes the back-electromotive force (Back-EMF) variations and armature resistance thermal dependencies.

Since the experimental platform lacks integrated real-time physical current sensors on its embedded hardware, a virtual current sensing loop was developed in software. To establish this sensorless framework, a preliminary steady-state empirical characterization was executed to map the relationship between the applied control variable (PWM) and the resulting armature current (I).

The experimental curves obtained within the full operational envelope revealed a highly linear correlation for both actuators in steady-state, mathematically modeled as:

$$I(t) = a \cdot PWM(t) + b \quad (17)$$

where a and b represent the empirical slopes and offsets, respectively. The identified parametric functions for each propulsion unit are summarized in Table 2.

Table 2: Identified PWM-to-Current calibration functions

Actuator	Calibration Function $I(t) = a \cdot PWM(t) + b$
Front	$I_1(t) = 0.006427 \cdot PWM_1(t) - 7.139481$
Rear	$I_2(t) = 0.007061 \cdot PWM_2(t) - 7.643550$

For the physical embedded implementation, this mapping is inverted inside the processing unit to establish the virtual current loop. The software execution framework processes the demanded reference current (I_{ref}) for each motor and maps it into a corresponding digital command through the following static inversion sequence:

1. The algorithm takes the desired current variable, $I_{\text{ref}}(t)$, which acts as the intermediate control input for the actuation subsystem.
2. The software execution loop applies the analytical inverse of the experimentally identified linear functions to convert this reference current into its corresponding digital PWM duty

cycle:

$$\text{PWM}(t) = \frac{I_{\text{ref}}(t) - b}{a} \quad (18)$$

where a and b represent the specific empirical parameters of the respective actuator, as consolidated in Table 2.

3. The computed duty cycle is directly dispatched to the microprocessor internal timers to synthesize the physical hardware signal that drives the ESCs.

This procedure implements a virtual current loop entirely in software, simulating the behavior of a physical current-driven actuator without the need for real-time sensing infrastructure.

This methodology successfully implements an indirect virtual current loop in software, replicating the performance of a cascaded hardware control topology without requiring real-time physical current sensing infrastructure.

5 Control system design

6 Results and discussion

7 Conclusion and future work

————— daqui para baixo é do template, não apaguei porque estou usando a estrutura para table and equations —————

8 Equations

Equations in $\text{\LaTeX}$ can either be inline or on-a-line by itself (“display equations”). For inline equations use the $\$...\$$ commands. E.g.: The equation $H\psi = E\psi$ is written via the command $\$H \backslash\text{psi} = E \backslash\text{psi}\$$.

For display equations (with auto generated equation numbers) one can use the equation or align environments:

$$\|\tilde{X}(k)\|^2 \leq \frac{\sum_{i=1}^p \|\tilde{Y}_i(k)\|^2 + \sum_{j=1}^q \|\tilde{Z}_j(k)\|^2}{p+q}. \quad (19)$$

where,

$$\begin{aligned} D_\mu &= \partial_\mu - ig \frac{\lambda^a}{2} A_\mu^a \\ F_{\mu\nu}^a &= \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc} A_\mu^b A_\nu^c \end{aligned} \quad (20)$$

Notice the use of $\backslash\text{nonumber}$ in the align environment at the end of each line, except the last, so as not to produce equation numbers on lines where no equation numbers are required. The $\backslash\text{label}\{\}$ command should only be used at the last line of an align environment where $\backslash\text{nonumber}$ is not used.

$$Y_\infty = \left(\frac{m}{\text{GeV}}\right)^{-3} \left[1 + \frac{3 \ln(m/\text{GeV})}{15} + \frac{\ln(c_2/5)}{15}\right] \quad (21)$$

The class file also supports the use of $\backslash\text{mathbb}\{\}$, $\backslash\text{mathscr}\{\}$ and $\backslash\text{mathcal}\{\}$ commands. As such $\backslash\text{mathbb}\{\mathbb{R}\}$, $\backslash\text{mathscr}\{\mathcal{R}\}$ and $\backslash\text{mathcal}\{\mathcal{R}\}$ produces $\mathbb{R}$, $\mathscr{R}$ and $\mathcal{R}$ respectively (refer Subsubsection ??).

9 Tables

Tables can be inserted via the normal table and tabular environment. To put footnotes inside tables you should use $\backslash\text{footnotetext}\{\dots\}$ tag. The footnote appears just below the table itself (refer Tables 3 and 4). For the corresponding footnote mark use $\backslash\text{footnotemark}\{\dots\}$

Table 3: Caption text

Column 1	Column 2	Column 3	Column 4
row 1	data 1	data 2	data 3
row 2	data 4	data 5 ¹	data 6
row 3	data 7	data 8	data 9 ²

Source: This is an example of table footnote. This is an example of table footnote.

¹Example for a first table footnote. This is an example of table footnote.

²Example for a second table footnote. This is an example of table footnote.

The input format for the above table is as follows:

```
\begin{table}[<placement-specifier>]
\caption{<table-caption>}\label{<table-label>}%
\begin{tabular}{@{}llll@{}}
\toprule
Column 1 & Column 2 & Column 3 & Column 4\\
\midrule
row 1 & data 1 & data 2 & data 3 \\
row 2 & data 4 & data 5\footnotemark[1] & data 6 \\
```

```

row 3 & data 7 & data 8 & data 9\footnotemark[2]\\
\botrule
\end{tabular}
\footnotetext{Source: This is an example of table footnote.
This is an example of table footnote.}
\footnotetext[1]{Example for a first table footnote.
This is an example of table footnote.}
\footnotetext[2]{Example for a second table footnote.
This is an example of table footnote.}
\end{table}

```

Table 4: Example of a lengthy table which is set to full textwidth

Project	Element 1 ¹			Energy	Element 2 ²		
	Energy	σ_{calc}	σ_{expt}		Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 ± 12	990 A	780 A	1166	1239 ± 100
Element 4	500 A	961	922 ± 10	990 A	900 A	1268	1092 ± 40
				990 A	780 A	1166	1239 ± 100
					900 A	1268	1092 ± 40

Note: This is an example of table footnote. This is an example of table footnote this is an example of table footnote this is an example of table footnote.

¹Example for a first table footnote.

²Example for a second table footnote.

In case of double column layout, tables which do not fit in single column width should be set to full text width. For this, you need to use `\begin{table*} ... \end{table*}` instead of `\begin{table} ... \end{table}` environment. Lengthy tables which do not fit in textwidth should be set as rotated table. For this, you need to use `\begin{sidewaystable} ... \end{sidewaystable}` instead of `\begin{table*} ... \end{table*}` environment. This environment puts tables rotated to single column width. For tables rotated to double column width, use `\begin{sidewaystable*} ... \end{sidewaystable*}`.

10 Figures

As per the L^AT_EX standards you need to use eps images for L^AT_EX compilation and pdf/jpg/png images for PDFL^AT_EX compilation. This is one of the major difference between L^AT_EX and PDFL^AT_EX. Each image should be from a single input .eps/vector image file. Avoid using subfigures. The command for inserting images for L^AT_EX

Table 5: Tables which are too long to fit, should be written using the `\begin{sidewaystable} ... \end{sidewaystable}` environment as shown here

Projectile	Element 1 ¹			Energy	Element 2 ²		
	Energy	σ_{calc}	σ_{expt}		Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 ± 12	990 A	780 A	1166	1239 ± 100
Element 4	500 A	961	922 ± 10	990 A	900 A	1268	1092 ± 40
Element 5	990 A	1168	1547 ± 12	990 A	780 A	1166	1239 ± 100
Element 6	500 A	961	922 ± 10	990 A	900 A	1268	1092 ± 40

Note: This is an example of table footnote this is an example of table footnote this is an example of table footnote this is an example of table footnote.

¹This is an example of table footnote.

and PDFLaTeX can be generalized. The package used to insert images in LaTeX/PDFLaTeX is the graphicx package. Figures can be inserted via the normal figure environment as shown in the below example:

```
\begin{figure}[<placement-specifier>]
\centering
\includegraphics{<eps-file>}
\caption{<figure-caption>}\label{<figure-label>}
\end{figure}
```



Fig. 3: This is a widefig. This is an example of long caption this is an example of long caption this is an example of long caption this is an example of long caption

In case of double column layout, the above format puts figure captions/images to single column width. To get spanned images, we need to provide `\begin{figure*} ... \end{figure*}`.

For sample purpose, we have included the width of images in the optional argument of `\includegraphics` tag. Please ignore this.

11 Algorithms, Program codes and Listings

Packages `algorithm`, `algorithmicx` and `algpseudocode` are used for setting algorithms in LaTeX using the format:

```
\begin{algorithm}
\caption{<alg-caption>}\label{<alg-label>}
\begin{algorithmic}[1]
. . .
\end{algorithmic}
\end{algorithm}
```

You may refer above listed package documentations for more details before setting `algorithm` environment. For program codes, the “verbatim” package is required and the command to be used is `\begin{verbatim} ... \end{verbatim}`.

Similarly, for listings, use the `listings` package. `\begin{lstlisting} ... \end{lstlisting}` is used to set environments similar to `verbatim` environment. Refer to the `lstlisting` package documentation for more details.

A fast exponentiation procedure:

```
begin
  for i := 1 to 10 step 1 do
    expt(2, i);
  newline() od
will be set flush to the right margin
where
proc expt(x, n) ≡
  z := 1;
  do if n = 0 then exit fi;
  do if odd(n) then exit fi,
    comment: This is a comment statement;
    n := n/2; x := x * x od;
  { n > 0 };
  n := n - 1; z := z * x od;
  print(z).
end
```

Comments

Algorithm 1 Calculate $y = x^n$

Require: $n \geq 0 \vee x \neq 0$

Ensure: $y = x^n$

```
1:  $y \leftarrow 1$ 
2: if  $n < 0$  then
3:    $X \leftarrow 1/x$ 
4:    $N \leftarrow -n$ 
5: else
6:    $X \leftarrow x$ 
7:    $N \leftarrow n$ 
8: end if
9: while  $N \neq 0$  do
10:  if  $N$  is even then
11:     $X \leftarrow X \times X$ 
12:     $N \leftarrow N/2$ 
13:  else [ $N$  is odd]
14:     $y \leftarrow y \times X$ 
15:     $N \leftarrow N - 1$ 
16:  end if
17: end while
```

```

for i:=maxint to 0 do
begin
  { do nothing }
end;
Write('Case-insensitive-');
Write('Pascal-keywords.');
```

12 Cross referencing

Environments such as figure, table, equation and align can have a label declared via the `\label{#label}` command. For figures and table environments use the `\label{}` command inside or just below the `\caption{}` command. You can then use the `\ref{#label}` command to cross-reference them. As an example, consider the label declared for Figure 3 which is `\label{fig1}`. To cross-reference it, use the command `\ref{fig1}`, for which it comes up as “Figure 3”.

To reference line numbers in an algorithm, consider the label declared for the line number 2 of Algorithm 1 is `\label{algn2}`. To cross-reference it, use the command `\ref{algn2}` for which it comes up as line 2 of Algorithm 1.

12.1 Details on reference citations

Standard L^AT_EX permits only numerical citations. To support both numerical and author-year citations this template uses `natbib` L^AT_EX package. For style guidance please refer to the template user manual.

Here is an example for `\cite{...}`: [?]. Another example for `\citep{...}`: [?]. For author-year citation mode, `\cite{...}` prints Jones et al. (1990) and `\citep{...}` prints (Jones et al., 1990).

All cited bib entries are printed at the end of this article: [?], [?], [?], [?], [?], [?], [?], [?], [?], [?] and [?].

13 Examples for theorem like environments

For theorem like environments, we require `amsthm` package. There are three types of predefined theorem styles exists—`thmstyleone`, `thmstyletwo` and `thmstylethree`

<code>thmstyleone</code>	Numbered, theorem head in bold font and text in italic style
<code>thmstyletwo</code>	Numbered, theorem head in roman font and text in italic style
<code>thmstylethree</code>	Numbered, theorem head in bold font and text in roman style

For mathematics journals, theorem styles can be included as shown in the following examples:

Theorem 1 (Theorem subhead) *Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Proposition 2 *Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Example 1 Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem.

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Remark 1 Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia.

Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem.

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Definition 1 (Definition sub head) Example definition text. Example definition text. Example definition text. Example definition text. Example definition text. Example definition text. Example definition text.

Additionally a predefined “proof” environment is available: `\begin{proof} ... \end{proof}`. This prints a “Proof” head in italic font style and the “body text” in roman font style with an open square at the end of each proof environment.

Proof Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. □

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Proof of Theorem 1 Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. □

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Appendix A Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.